

Model Analysis Report

Wind Damage Model

MKM-WD-001

Report Date: 21 July 2026

Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Pure vulnerability curve: damage ratio by peak sustained wind, at the configured base $v_{50} = 27.8$ m/s and slope $a = 0.2$ m⁻¹s.

Peak wind (m/s)	Damage ratio
0.0000	0.0000
10.0	0.0277
17.5	0.1130
22.0	0.2387
27.8	0.5000
33.0	0.7389
40.0	0.9198
50.0	0.9883
60.0	0.9984
70.0	0.9998

Table 2: Signed v_{50} shift (m/s) by BRI wind sub-score and composite score, capped at ± 12.0 m/s. Positive shifts move the curve rightward (the building tolerates more wind before damage).

BRI wind score	Composite NR (0.190)	Composite B (0.500)	Composite A (0.745)	Comp
0.1900	-8.71	-5.81	-4.61	
0.5000	-2.90	0.0000	1.20	
0.7450	-0.5100	2.39	3.59	
0.9350	0.8530	3.76	4.95	

Table 3: BRI-adjusted damage ratio by resilience grade at a common operational threshold of 27.8 m/s (wind sub-score and composite both set to the grade mid-point).

Peak wind (m/s)	NR grade	B grade	A grade	AA grade
0.0000	0.0000	0.0000	0.0000	0.0000
10.0	0.1396	0.0277	0.0137	0.0091
17.5	0.4211	0.1130	0.0585	0.0397
22.0	0.6414	0.2387	0.1326	0.0922
27.8	0.8509	0.5000	0.3279	0.2448
33.0	0.9417	0.7389	0.5799	0.4783
40.0	0.9850	0.9198	0.8484	0.7881
50.0	0.9979	0.9883	0.9764	0.9649
60.0	0.9997	0.9984	0.9967	0.9951
70.0	1.00	0.9998	0.9996	0.9993

Table 4: Persistence multiplier $\Phi(t)$ by hours above the damage-onset threshold ($\phi_{gust} = 0.45$, $\tau = 4.0$ h). A momentary gust realises only the floor; a prolonged blow saturates at the full peak-curve damage.

Duration above threshold (h)	$\Phi(t)$
0.0000	0.4500
0.5000	0.5146
1.00	0.5717
2.00	0.6664
4.00	0.7977
6.00	0.8773
8.00	0.9256
12.0	0.9726
24.0	0.9986

Table 5: Persistence-aware damage ratio (severity $\times$ duration) for a B-grade building at a 27.8 m/s operational threshold.

Peak wind (m/s)	$t = 0$ h	$t = 1$ h	$t = 2$ h	$t = 4$ h	$t = 8$ h	$t = 24$ h
22.0	0.1074	0.1364	0.1590	0.1904	0.2209	0.2383
27.8	0.2250	0.2858	0.3332	0.3988	0.4628	0.4993
33.0	0.3325	0.4224	0.4924	0.5894	0.6839	0.7378
40.0	0.4139	0.5258	0.6130	0.7337	0.8514	0.9186
50.0	0.4448	0.5650	0.6586	0.7884	0.9148	0.9870
60.0	0.4493	0.5707	0.6653	0.7964	0.9241	0.9970

Table 6: PRS damage-onset trigger by resilience grade. The trigger fires when the peak sustained wind reaches the BRI-adjusted effective threshold, so a resilient building fires less readily at the same wind.

Grade	$v_{50,eff}$ (m/s)	30 m/s	35 m/s	40 m/s	45 m/s
NR	19.1	yes	yes	yes	yes
B	27.8	yes	yes	yes	yes
A	31.4	no	yes	yes	yes
AA	33.4	no	yes	yes	yes

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 7: Stress testing scenarios for Wind Damage Model.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2σ inputs	Medium	Pending
Severe stress	+3σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model’s validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 8: Performance metrics for Wind Damage Model.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.